MIL-S-19500/257A(EL)
6 February 1968

SUPERSEDING
MIL-S-19500/257(EL)
15 October 1962

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON
TYPE 1N658

1. SCOPE

1.1 Scope.- This specification covers the detail requirements for a silicon, semiconductor diode for use, particularly, as a switching device in relatively high-current-load equipment circuits.

1.2 Outline and dimensions.- See Fig. 1 herein.

1.3 Operating characteristics and ratings.- (At T_A = +25°C, unless otherwise specified. See 3.4 herein.)

	V_R	I_o	i_f(surge at: 1/120 sec.)	P	Switching (t_{rr})	$T_{(oper.)}$	T_{stg}	Barom. Press.
	Vdc	mAdc	mAdc	mW	usec	°C	°C	(alt.)ft.
Min.	---	---	---	---	---	-65	-65	---
Max.	-120	200	600	200	0.30	+150	+175	85,000

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS
MILITARY
MIL-S-19500 — Semiconductor Devices, General Specification For

STANDARDS
MILITARY
MIL-STD-15 — Electrical And Electronics Symbols
MIL-STD-750 — Test Methods For Semiconductor Devices

FSC-5961

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Requirements.- Requirements for the diodes shall be in accordance with Specification MIL-S-19500 and as otherwise specified herein.

3.2 Abbreviations and symbols.- The abbreviations and symbols used herein are defined Specification MIL-S-19500, and as follows:

C_L . . . . . load capacitance
R_L . . . . . load resistance
R_r . . . . . inverse recovery resistance

3.3 Design and construction.- The diode shall be of the design, construction, and physical dimensions specified in Figure 1.

3.3.1 Terminal arrangement.- The terminal arrangement on the diode shall be as indicated in Figure 1.

3.3.2 Operating position.- The diode shall be capable of proper operation in any position.

3.4 Performance characteristics.- The diode performance characteristics shall be as specified in Tables I and II herein.

3.5 Marking.- Except as otherwise specified herein, marking shall be in accordance with Specification MIL-S-19500. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the "classification" type designation only. The "Manufacturer's identification" and "country of origin" may, at option of the manufacturer, be omitted from being marked directly on the semiconductor device covered herein. (See 3.5.1 herein.)

3.5.1 Polarity marking.- The graphic symbol for polarity indication on the semiconductor diode shall be as designated in Standard MIL-STD-15.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Except as otherwise specified herein, the responsibility for inspection, general procedures for acceptance, classification of inspection, and inspection conditions and methods of test shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions.

4.2 Qualification and acceptance inspection.- Qualification and Quality Conformance inspection shall be in accordance with Specification MIL-S-19500, Quality Assurance Provisions, and as otherwise specified herein. Groups A and B inspection shall consist of the examinations and tests specified in Tables I and II, respectively, herein. Quality Conformance inspection shall include inspection of Preparation for Delivery (see 5.1 herein).

4.2.1 Specified LTPD for subgroups.- The LTPD specified for a subgroup in Tables I and II herein shall apply for all of the tests, combined, in the subgroup.

4.2.2 Dispostion of sample units.- Sample units that have been subjected to and have passed Group B, Subgroups 1 through 8 tests (these tests to be considered non-destructive), may be delivered on the contract or order provided that, after Group B inspection is terminated, those sample units are subjected to and pass Group A inspection. Defective units from any sample group that may have passed group inspection shall not be delivered on the contract or order until the defect(s) has been remedied to the satisfaction of the Government.

4.3 Particular examination and test requirements.-

4.3.1 Interval for End-Point test measurements.- All applicable End-Point Test measurements shall be performed, after sample units have been subjected to required physical-mechanical or environmental test(s), in accordance with the following time-delay limitations:

(a) For Qualification inspection: within 24 hours.

(b) For Quality Conformance inspection: within 96 hours.

4.3.2 Mechanical damage resulting from tests.- Except for intentionally deforming, mutilating, or dismembering mechanical-stress tests to which samples are subjected, there shall be no evidence of mechanical damage to any sample unit as a result of any of the Group A or B tests.

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Table I. Group A inspection.

Test Method per MIL-STD-750	Examination or test	Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		5				
2071	Visual and mechanical examination	---		---	---	---	---
	Subgroup 2		5				
4011	Forward voltage drop	I_F = 100 mAdc		V_F	---	1.0	Vdc
4016	Reverse current	V_R = -50 Vdc		I_R	---	0.05	uAdc
4016	Reverse current	V_R = -120 Vdc		I_R	---	5.0	uAdc
4041	Rectification efficiency	Test Cond.A V = 40 Vac f = 60 Hz R_L = 600 ohms		I_o	28	---	mAdc
4031	Reverse recovery time	Test Cond.A I_F = 5 mAdc V_R = -40 Vdc R_L = 2,000 ohms, ± 10% C_L = 10 pf f = 100 Hz R_r = 80 k ohms		t_{rr}	---	0.30	usec
1/	High-temperature operation:	T_A = +150° $^{+5°}_{-0°}$ C					
4016	Reverse current	V_R = -50 Vdc		I_R	---	25	uAdc

1/ Test measurement shall be made after thermal equilllibrium has been reached at the temperature specified.

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Table II. Group B inspection.

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
	Subgroup 1		15				
2066	Physical dimensions	---		---	---	---	---
	Subgroup 2		20				
1051	Temperature cycling	Test Cond.F		---	---	---	---
1056	Thermal shock (glass strain)	Test Cond.A		---	---	---	---
1021	Moisture resistance	No initial conditioning		---	---	---	---
	End-point tests:						
4011	Forward voltage drop	I_F = 100 mAdc		V_F	---	1.0	Vdc
4016	Reverse current	V_R = -50 Vdc		I_R	---	0.05	uAdc
	Subgroup 3		20				
2016	Shock	Non-operating 500 G 5 blows of 1 msec ea. in orientations Y1, Y2, Z1 (total = 15 blows)		---	---	---	---
2006	Constant acceleration	20,000 G Orientations Y1, Y2, Z1		---	---	---	---
2046	Vibration fatigue	Non-operating 10 G Orientations Y1, Z1		---	---	---	---
2056	Vibration, variable frequency	10 G Orientations Y1, Z1		---	---	---	---
	End-point tests:						
	Same as for Subgroup 2, above						

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Table II. Group B inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
	Subgroup 4		20				
2036	Terminal strength: Tension	Test Cond.A Weight = 2 lbs Time = 3 seconds		---	---	---	---
	Lead fatigue	Test Cond.E Weight = 1 lb. Max.time ea.bend = 1 minute		---	---	---	---
	End-point tests: Same as for Subgroup 2, above						
	Subgroup 5		20				
2/	Low-temperature operation:	T_C = -65 $^{+0°}_{-5°}$C					
4011	Forward voltage drop	I_F = 10 mAdc		V_F	---	1.0	Vdc
4016	Reverse current	V_R = -50 Vdc		I_R	---	0.05	uAdc
1001	Barometric pressure, reduced (altitude operation)	Pressure = 15 mmHg Energization duration = 1 minute v_r = 125 v					
1046	Salt spray(corrosion)	Test Cond.A		---	---	---	---
	End-point tests: Same as for Subgroup 2, above						

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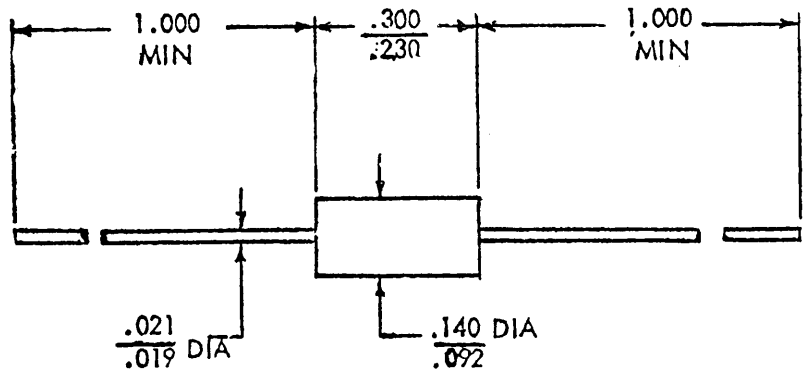
Table II. Group B inspection-(Cont'd).

Test Method per MIL-STD-750	Examination or test 1/	Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
	Subgroup 6		20				
4066	Surge current	i_f(surge)=600 mAdc T_C = + 100° +5°C -0° 10 surges at 1- min. intervals v_r = 100 v		---	---	---	---
	End-point tests:						
	Same as for Subgroup 2, above						
	Subgroup 7		λ=10				
1031	High-temperature life (non-operating)	T_{stg}= +175° +5°C -0°C		---	---	---	---
	End-point tests:						
	Same as for Subgroup 2, above						
	Subgroup 8		λ=10				
1036	Intermittent operation	Operation = 100 min. on;20 min. off T_A = 25°C;f = 60 Hz R_L = 112 ohms I_o = 200 mAdc		---	---	---	---
	End-point tests:						
	Same as for Subgroup 2, above						

1/ See 4.3.1 herein.

2/ Test measurement(s) shall be made after thermal equilibrium has been reached at the temperature specified.

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All dimensions in inches

Figure 1. Outline and dimensions.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery.- Preparation for delivery, and the inspection of preparation for delivery, shall be in accordance with Specification MIL-S-19500.

6. NOTES

6.1 Notes.- The notes included in Specification MIL-S-19500, with the following additions or exceptions, are applicable to this document.

6.2 Application guidance.- The revised requirements in this document issue do not affect replaceability between the diodes covered herein and the diodes covered in the previous issue of this document.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List (QPL)-19500, whether or not such products have actually been so listed by that date. Information pertaining to qualification of products covered by this specification should be requested from the Commanding General, U. S. Army Electronics Command, Fort Monmouth, New Jersey 07703, Attention: AMSEL-PP-EM-2.

6.4 Revision (document) changes.- Revision-letter symbols are not used in this document revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
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Preparing activity:
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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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